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EVOLUTION

Fossil evidence favors a role for vision in the modular evolution of the primate neocortex

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Neocortical expansion is a key innovation of crown primates, although fossil evidence for how the neocortex has evolved remains elusive. We studied neocortical evolution using virtual brain endocasts from extant and Eocene-Miocene fossil primates. We found that neocortex size increased in parallel across crown primate clades, with rapid increases in tarsiers and anthropoids. Although the frontal lobe is commonly described as having expanded disproportionately in primates, we show that it evolved gradually, following a common pattern for allometric scaling. By contrast, nonfrontal neocortical regions underwent rapid increases relative to brain size in haplorhines. Neocortical expansion occurred in concert with increases in optic foramen area, suggesting that visual sensory inputs are functionally tied to modular evolution of the primate neocortex.

Many primates—including humans, aye-ayes, and various monkeys and apes—evolved a larger brain than that of most other mammals of comparable body mass (*1*). Fossil brain endocasts show that brain size has substantially increased relative to body size in every major extant primate lineage (*1*–*3*). The large brains of primates reflect disproportionate expansion of the cerebrum (*4*, *5*), with the primate cerebral cortex comprising a larger percentage of brain mass than in other mammals, and the neocortex in particular is greatly expanded (*6*). Substantial variation in neocortex size relative to brain size is also evident within primates, in which haplorhines have a larger neocortex than that of strepsirrhines (*7*–*9*). Neocortical expansion in primates is thought to be closely tied to changes in the visual system and an elaboration of cortical visual regions (*10*, *11*). In primates, large portions of the temporal, parietal, and (especially) occipital regions of the neocortex are devoted to processing visual inputs, and neocortical regions devoted to processing auditory and somatosensory inputs are proportionally much smaller (*12*–*14*). Although the frontal lobe also receives inputs from extrastriate visual areas (*15*), the frontal neocortex in primates is less specialized for processing visual information (*16*). Therefore, if neocortical expansion in primates is closely tied to evolutionary changes in the visual system, then the nonfrontal neocortex would be expected to have expanded disproportionately relative to the frontal neocortex.

Despite having a smaller neocortex relative to body mass and brain volume (*8*, *9*, *17*), extant strepsirrhines resemble haplorhines in the scaling of frontal neocortex relative to nonfrontal neocortex. The proportion of frontal gray matter volume to nonfrontal gray matter volume is broadly similar in haplorhines and strepsirrhines, and the two variables are tightly and log-linearly correlated across primates (*18*). Nevertheless, claims persist in the paleontological literature that frontal lobe size increases in many primate lineages from the Eocene onward (*19*–*23*). These claims are based on gross observation of endocranial shape and have not been tested quantitatively.

In contrast to the frontal lobe scaling relationship shared across living primates, haplorhines and strepsirrhines differ in other key aspects of their neocortical anatomy and scaling. Relative to brain size, haplorhines have a larger nonfrontal neocortex than that of strepsirrhines [data are available in (*18*)]. Within the occipital lobe, the primary visual cortex (V1) is larger in extant haplorhines than in strepsirrhines (*3*, *8*, *9*, *17*).

We provide a broad comparative analysis for the evolution of neocortical proportions in a dataset that combines extant and fossil euarchontans (Primates, Dermoptera, Scandentia, and Plesiadapiformes) through quantification of frontal and nonfrontal portions of the neocortical surface of virtual endocasts. Fossil endocasts provide the only direct evidence for the timing and sequence of primate brain evolution and allow extinct taxa to be incorporated into comparative analyses. Using these data, we reconstructed ancestral states for brain size, neocortex size, and neocortical subregions across euarchontans and estimated evolutionary rate shifts over the past ~56 million years. We tested whether frontal lobe expansion evolved independently in multiple primate clades as proposed in previous studies. Using optic foramen size as an osteological proxy for visual input to the brain (*24*, *25*), we also examined the role of visual input on neocortical evolution in primates.

Results

The phylogeny used here (Fig. 1) is based on morphological and molecular data [supplementary materials (SM) SD 4] (*26*–*32*). Using virtual endocasts, we measured endocast volume (ECV) and the exposed surface area of the neocortex (NSA). In crown primates and tupaiids, we measured the neocortical surface area of the frontal lobe (FSA), which includes both prefrontal and primary motor cortex, and the summed surface areas of the occipital, parietal, and temporal lobes (OPT) (data S1). The volume and surface measurements from endocasts reliably estimate the known sizes of these variables reported in histological studies (validation is modeled in SM SD2; details of Bayesian tests in Results are elaborated in table S1). The optic foramen transmits the optic nerve and is a reliable proxy for the total amount of visual input to the brain in primates (*24*, *25*, *33*).

Neocortex size evolution

Substantial increases in relative neocortex size occurred with the origin of crown primates and within each of the major crown primate clades (*1*, *34*). In a plot of NSA relative to total endocast volume, the Early Eocene plesiadapiform *Ignacius* plots within the envelope of extant scandentians and dermopterans and below that of crown primates (Fig. 2A). *Ignacius* thus groups outside of crown primates in relative neocortex size (*35*), which is supported by a Bayes factor (BF) of 7. Crown strepsirrhines have a larger NSA relative to endocast volume than that of dermopterans and scandentians [Bayesian *P* value (Bp) = 0.006]. In turn, living haplorhines have a significantly larger NSA than that of strepsirrhines of comparable endocranial volume (Bp < 0.0001) (Fig. 2A). Further changes in the relative size of the neocortex occurred independently in various lineages (Figs. 2A and 3A) (*1*): Crown strepsirrhine NSA enlarged over that of their Eocene sister group, the Adapoidea (Bp = 0.008) (*2*); Omomyoidea fall within the range of crown strepsirrhines and have relatively smaller neocortices than those of tarsiers and anthropoids (Bp < 0.0001) (*3*). The neocortices of tarsiers and anthropoids are substantially larger than those of adapoids, crown strepsirrhines, and omomyoids of comparable endocranial volume (Bp < 0.0001) (Fig. 2A). A variable-rates regression model of NSA controlling for endocranial volume (Fig. 3A) is overwhelmingly favored over a gradually evolving model (BF = 18), with the largest rate shifts

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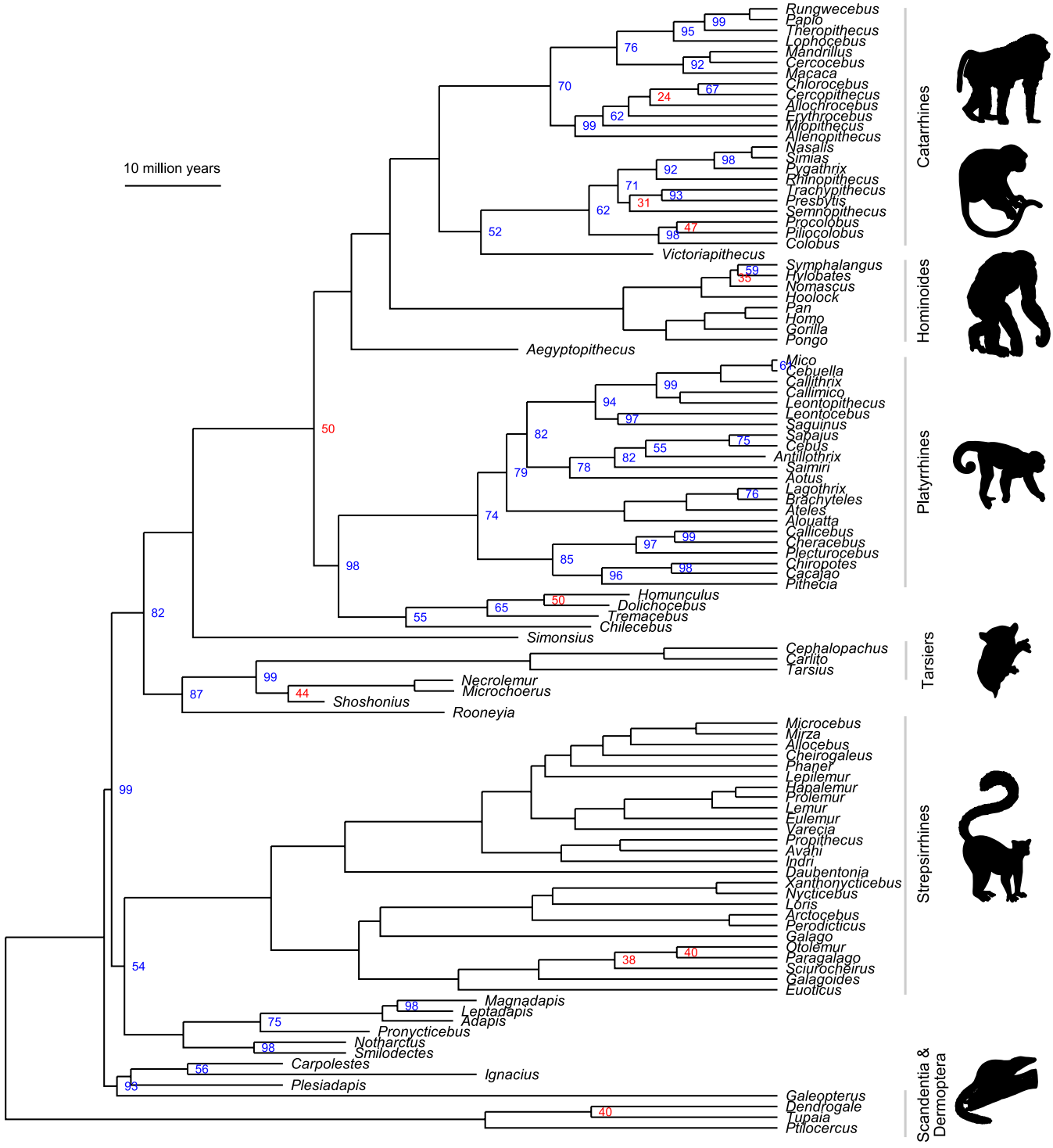


Fig. 1. Euarchontan phylogeny inferred from morphological and molecular data. Analysis was performed by using MrBayes (v3.2.7). The total dataset contains 137 taxa and 6741 characters (399 standard and 6342 DNA). Node values indicate Bayesian posterior support. [Silhouettes are from https://www.PhyloPic.org. Credits: *Cebus* silhouette, Sarah Werning, CC-BY-3.0; *Lemur catta* by Roberto Díaz Sibaja, CC-BY-3.0]

associated with a relative increase in NSA in the stem anthropoid [median correlation coefficient (r) = 75×] and stem tarsiid (median r = 11×) lineages (Fig. 3A). Rate variability of NSA evolution is also large in adapoids (Fig. 3A), but when we removed adapoids from the analyses, we still found strong evidence for variable rates of evolution in NSA across primates (BF = 15).

Anthropoids have exhibited relatively large neocortices compared with those of other primate radiations since at least the early Oligocene. Enlarged neocortices comparable with those of extant anthropoids are already evident in the early Oligocene stem anthropoid *Simonsius* and the stem catarrhine *Aegyptopithecus* (Fig. 2A). The neocortex is also enlarged in the stem platyrrhines *Homunculus* and *Tremacebus* (*3*).

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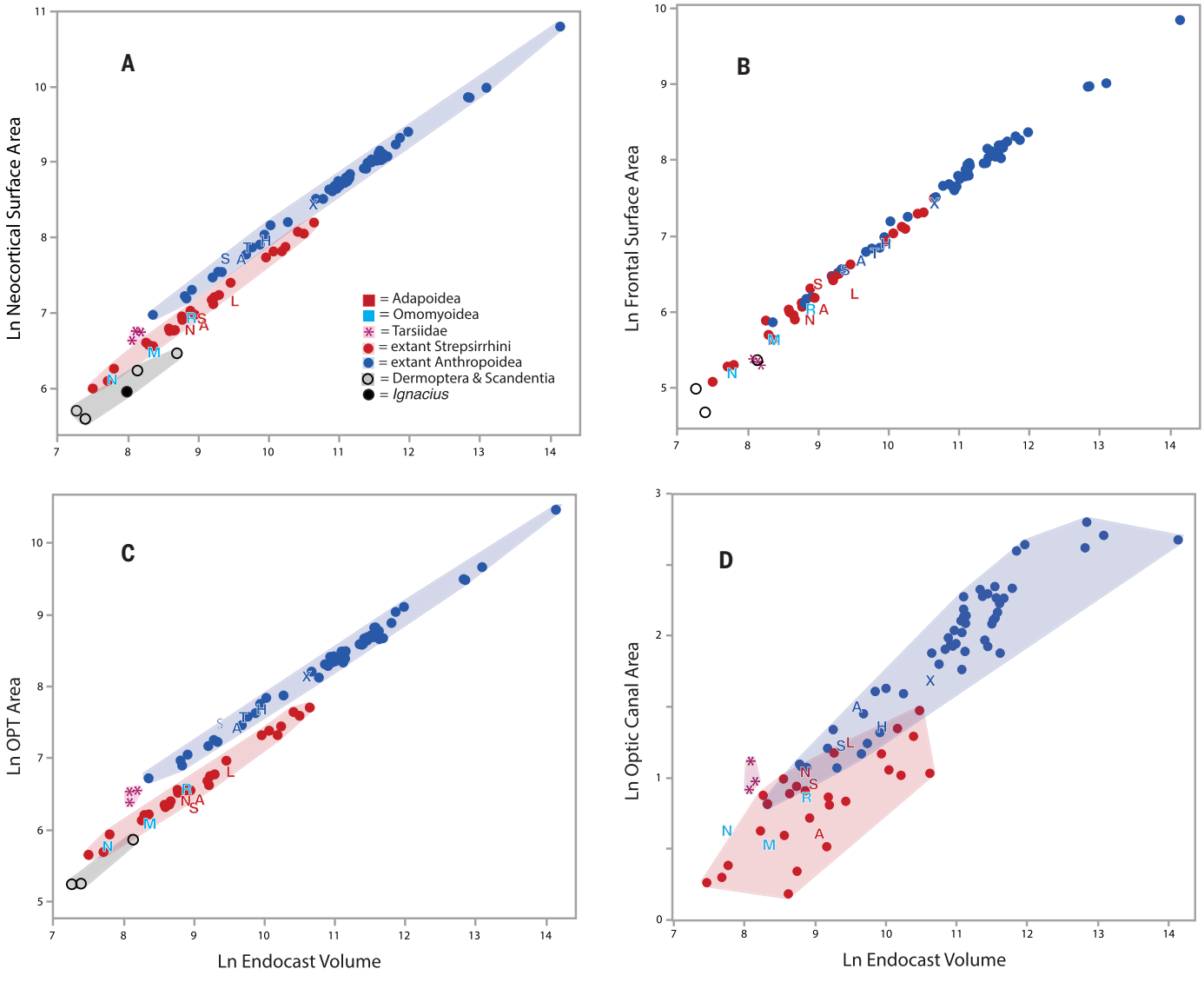


Fig. 2. Bivariate plots of brain and orbit surface areas compared with endocast volume. (**A**) Ln neocortical surface area compared with Ln endocast volume. (**B**) Ln frontal lobe surface area compared with Ln endocast volume. (**C**) Ln OPT surface area compared with Ln endocast volume. (**D**) Ln optic canal cross-sectional area compared with Ln endocranial volume. Fossil taxa are indicated with letters, with anthropoids in blue, omomyoids in cyan, and adapoids in red. Blue symbols indicate early Oligocene stem anthropoid *Simonsius grangeri* (S), early Oligocene stem catarrhine *Aegyptopithecus zeuxis* (A), Early Miocene stem platyrrhines *Tremacebus harringtoni* (T) and *Homunculus patagonicus* (H), and Pleistocene cebid *Antillothrix bernensis* (X). Cyan symbols indicate Eocene omomyoids *Rooneyia vaijensis* (R), *Microchoerus erinaceus* (M), and *Necrolemur antiquus* (N). Red symbols indicate Eocene adapoids *Adapis parisiensis* (A), *Leptadapis leenhardti* (L), *Notharctus tenebrosus* (N), and *Smilodectes gracilis* (S). In (D), OFAs of scandentians, dermopterans, and *Ignacius* are omitted (explanation is provided in SM SD3).

Our results favor the hypothesis that relative neocortex size independently increased in tarsiers and anthropoids, presumably because the phylogeny we used reconstructs omomyoids as stem tarsiids. If omomyoids are stem haplorhines rather than stem tarsiids as others have proposed (*27*, *36*–*41*), then an increase in relative neocortex size may be a synapomorphy of crown Haplorhini rather than the result of parallel evolution in Anthropoidea and Tarsiidae.

Neocortical proportions

Primate neocortical expansion relative to brain volume did not occur en bloc; instead, there is strong evidence for modular evolution, with frontal and nonfrontal subregions evolving at different rates in different clades (Figs. 2, B and C, and 3, B and C) (*42*). The tempo and mode of FSA evolution relative to endocast volume were broadly similar across most primate clades. In a bivariate plot of FSA versus ECV, primates and tupaiids fall along a single positively allometric trend [slope = 0.72; Bayesian 95% confidence interval (CI): 0.69 to 0.74] (Fig. 2B). Crown strepsirrhines and adapoids do not differ significantly from other primates (Bp = 0.35), and tarsiid and anthropoid FSAs are not proportionally larger than those of adapoids, crown strepsirrhines, or omomyoids (Bp = 0.47). Oligocene–Early Miocene stem and crown anthropoids in our sample resemble extant anthropoids in relative FSA (Fig. 2B).

A variable-rates regression model of FSA controlling for endocast volume is weakly favored over a gradually evolving model (BF = 2.7), but there are large, inferred rate changes in adapoids over a geologically short period of time. To test whether adapoid rate diversity is driving support for our variable-rates model, we removed them from the analysis and

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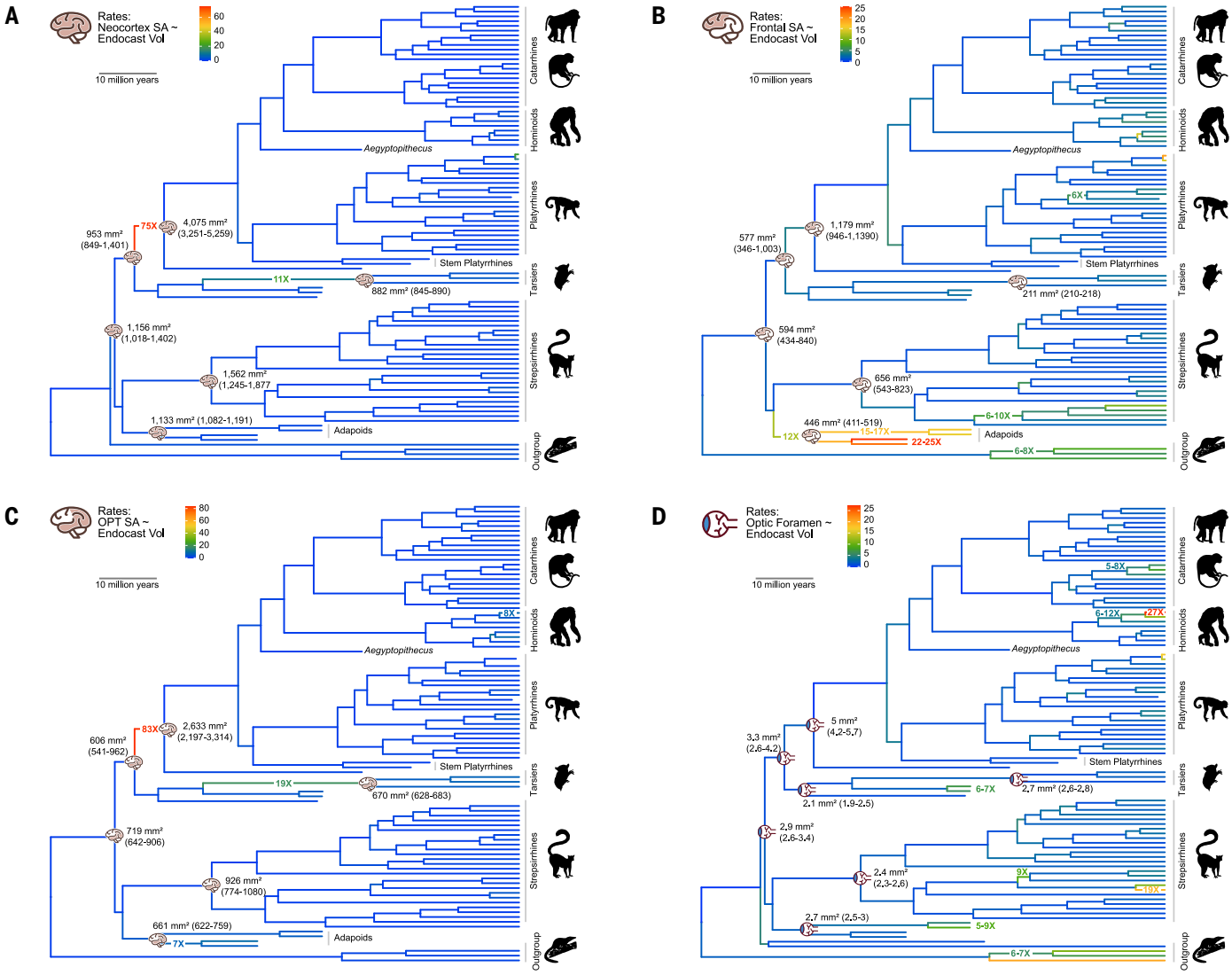


Fig. 3. Variable-rates regression models of Ln surface variables compared with Ln endocranial volume. (**A**) Neocortex surface area compared with endocranial volume. (**B**) Frontal lobe compared with endocranial volume. (**C**) OPT surface area compared with endocranial volume. (**D**) OFA compared with endocranial volume. These analyses sampled a posterior set of 2000 trees but are summarized on the MCC tree by using rate values for most recent common ancestors. [Silhouettes are from https://www.PhyloPic.org. Credits: *Cebus* silhouette, Sarah Werning, CC-BY-3.0; *Lemur catta* by Roberto Díaz Sibaja, CC-BY-3.0]

found support for gradual evolution of FSA across primates (BF = 4.9). Evidently, FSA evolved gradually as a function of endocast volume in most primate clades, with no large rate shifts in any anthropoid lineages, including Hominidae (Fig. 3B). Our ancestral state reconstruction of FSA in the last common ancestor (LCA) of primates broadly overlaps with that of the LCAs of adapoids and crown haplorhines (Fig. 3B).

Many have argued that the primate frontal lobe underwent an expansion in multiple lineages during anthropoid evolution (*43*–*45*). This conclusion was based partly on qualitative evaluation of fossil endocasts, with the Oligocene anthropoids *Simonsius* and *Aegyptopithecus* said to exhibit smaller frontal lobes compared with those of extant anthropoids (*22*, *23*). Similarly, qualitative analyses of endocast morphology in the Miocene stem platyrrhine *Chilecebus* (*22*) and stem cercopithecoid *Victoriapithecus* (*46*) have been used to suggest that frontal lobe expansion occurred independently in platyrrhines and catarrhines. However, our results demonstrate that fossil anthropoids follow the same positively allometric relationship of FSA relative to ECV (Fig. 2B) and body mass (*3*) observed in extant primates. Frontal lobe size in these early anthropoid taxa is thus entirely predictable on the basis of their brain size, and there is no evidence for disproportionate expansion in the size of the anthropoid frontal lobe since at least the early Oligocene.

In contrast to FSA, our data suggest that OPT has increased relative to endocast volume in multiple primate lineages (Figs. 2C and 3C). Taxa in our sample follow several parallel isometric slopes (median = 0.66, 95% Bayesian CI: 0.63 to 0.68) (Fig. 2C). Tupaiids have proportionally smaller OPTs than that of all living and extinct primates in our sample (Fig. 2C). Extant strepsirrhine OPTs are substantially larger than those of adapoids (Bp = 0.025). Eocene omomyoids (*Microchoerus*, *Necrolemur*, and *Rooneyia*) have OPTs that are more like those of crown strepsirrhines than those of tarsiers and anthropoids (Bp < 0.0001), with our model favoring a grouping of omomyoids with strepsirrhines by a BF of 33 for this trait. Tarsiids and anthropoids have relatively larger OPTs than those of omomyoids, adapoids, and crown strepsirrhines (Bp < 0.0001) (Fig. 2C). The greatly enlarged OPTs of

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extant anthropoids extend to fossil anthropoids at least as far back as the early Oligocene, with *Simonsius*, *Aegyptopithecus*, *Homunculus*, and *Tremacebus* OPTs falling squarely within the range of extant anthropoids of similar brain size (Fig. 2C).

Large evolutionary rate shifts in OPT surface area relative to endocast volume occurred in tarsiids and anthropoids, the two clades of primates that also have the largest neocortices and OPT surface areas for their brain size (BF = 16) (Fig. 2, A and C). Rate shifts associated with rapid enlargement of both the neocortex and its OPT component are found in the stem + crown anthropoid clade (median r = 83×) and in the lineage leading to crown tarsiids (median r = 19×) (Fig. 3C). Rate shifts in OPT surface area are also reconstructed for adapoids, but these rate changes are associated with projected decreases in relative neocortex size. With the removal of adapoids from the analyses, we still found strong evidence for variable rates of evolution in OPT size (BF = 11). Different rates of evolution of OPT size and FSA size shown in Fig. 3, B and C, are perhaps unsurprising considering evidence for substantial variability across mammals in rates of endocast shape evolution (*45*).

OPT and visual input

Evolutionary increase in relative neocortex and OPT size occur along the same primate lineage segments as the increase in visual input to the brain, as assessed from the optic foramen area (OFA) (*24*, *25*, *33*). Crown anthropoids and tarsiers have larger optic foramina than those of most crown strepsirrhines of similar endocranial volume (Bp = 0.045) (Fig. 2D). This association with large optic foramina is specific to the OPT and neocortex: Controlling for ECV in variable-rate models, the OFA is associated with OPT surface area (median slope = 0.35, 95% Bayesian CI: 0.06 to 0.67, Bp = 0.01) but not with FSA (median slope = −0.11, 95% Bayesian CI: −0.61 to 0.41, Bp = 0.33).

Among fossil taxa, omomyoids and adapoids have relatively small optic foramina resembling those of extant strepsirrhines (Bp = 0.29) (Fig. 2D). We reconstructed the LCA of living tarsiers as having evolved a larger optic foramen (median OFA = 2.7 mm^2, 95% CI = 2.6 to 2.8) than that of a hypothetical LCA of omomyoids and tarsiers (median OFA = 2.1 mm^2, 95% CI = 1.9 to 2.5). In parallel, we reconstructed the LCA of crown anthropoids + *Simonsius* as having a much larger optic foramen (median OFA = 5 mm^2, 95% CI = 4.2 to 5.7) than that of the LCA of extant haplorhines and omomyoids (median OFA = 3.3 mm^2, 95% CI = 2.6 to 4.2). Thus, the two clades that evolved much larger relative neocortex and OPT sizes compared with those of other euarchontans also have larger optic foramina and presumably have brains that process a substantially larger amount of visual input.

In this context, the phylogenetic position of omomyoids is crucial for determining whether increases in visual input and OPT size occurred once or twice during haplorhine evolution. If, as in our phylogenetic reconstruction, omomyoids are stem tarsiids, then increases in the amount of visual input and corresponding increases in the relative size of OPT and total neocortex size probably occurred in tarsiers and anthropoids in parallel. Alternatively, if omomyoids are stem haplorhines (*26*, *27*), then increases in visual input and relative OPT and neocortex size would likely have occurred only once in the haplorhine stem before the divergence of anthropoids and tarsiids.

As with NSA and OPT, we found that the OFA evolves at variable rates across primates (BF = 3.8, favoring a variable-rate model). However, the rate of optic foramen evolution shows a more heterogeneous pattern of evolution than do NSA or OPTs, with rate shifts occurring near the tips of the phylogenetic tree. Rate shifts are detected in treeshrews, strepsirrhines (*Arctocebus* and *Perodicticus*, median r = 11 to 19×), some colobines (median r = 5 to 8×), and hominids (median r = 6 to 27×, with the largest shift in the human lineage). These shifts appear to be driven by changes in brain size because a model of OFA alone evolves at uniform rates.

We did not identify large evolutionary rate changes in OFA relative to endocast size for either the anthropoid stem or the tarsiid stem (Fig. 3D). This result is perhaps less surprising for two reasons. First, if changes in OFA are partly responsible for changes in ECV, then one may expect to infer a more uniform rate of evolution when OFA is analyzed relative to ECV. Second, Bush and Allman have previously shown that small differences between anthropoid species in lateral geniculate nucleus (LGN) volume are associated with much larger differences in V1 size (*47*). These authors further speculated that extrastriate visual areas might also hyperscale relative to V1 size. If increases in visual input led to magnification of cortical representations of visual information in parallel processing streams outside of V1, then we expect smaller-magnitude changes in the absolute size of the optic foramen to be associated with larger-magnitude changes in the visual areas of the neocortex. For example, whereas neocortical and OPT surface areas are reconstructed as increasing 397 and 444%, respectively, from the crown haplorhine common ancestor to the anthropoid common ancestor, OFA is reconstructed as increasing a more modest 158% along this branch (Fig. 3). Such differences in the reconstructed absolute sizes of the structures considered in our study have predictable consequences for the reconstructed rates of evolution, with both NSA and OPTs evolving in the anthropoid stem at much higher rates than that of OFA (Fig. 3).

Conclusions

Our results suggest that modular evolution of neocortical size in primates is closely linked to the evolution of the visual system (*11*, *43*). As inferred and measured in this study, evolutionary increases in the size of the neocortex relative to brain size in tarsiers and anthropoids involved disproportionate expansion of occipital, temporal, and parietal areas and accompanying increases in visual input to the brain. We suggest that this association is not coincidental given that a large proportion of the nonfrontal neocortex in primates is occupied by functional areas that process visual information (*48*, *49*). Furthermore, our use of osteological metrics reveals that there also is an association between increased relative OPT size and visual input in a wide array of fossil stem and crown anthropoids. The shift to a larger proportion of the neocortex composed of the OPT regions thus appears to be an ancient feature of anthropoids dating at least to the early Oligocene, ~33 million years ago.

In contrast to the disproportionate expansion of OPT regions of the neocortex, our endocast measurements indicate that frontal lobe size has evolved along the same positively allometric trajectory relative to brain size in all major clades of primates throughout the Cenozoic. A similar relationship has been reported on the basis of measurements of frontal lobe tissues in living primates (*18*, *50*), suggesting that these findings may be tied to common developmental pathways in brain growth (*51*) rather than selection for expansion of the frontal cortex per se. We found no support for prior claims based on qualitative assessments of fossil endocasts that anthropoid brain evolution is marked by disproportionate expansion of the frontal lobes (and by inference in prefrontal regions) in multiple lineages independently (*22*, *23*, *46*).

Enlarged neocortices of anthropoids and tarsiers compared with those of strepsirrhines are thus primarily attributable to disproportionate expansion of occipital, temporal, and parietal regions of the neocortex and do not reflect disproportionate increases in frontal lobe size (*19*–*23*). Our results further suggest that expansion of OPT neocortex in haplorhines was favored by an increase in visual input to the cortex. Increased visual input in haplorhines is associated with the evolution of adaptations for higher visual acuity (*52*), including a retinal fovea (*24*), distinctive eye morphologies (*53*), and a postorbital partition that insulates the eye from mechanical disturbance (*54*). These adaptations may be the product of selection for improved visual functionality in a variety of contexts, such as social communication (*55*).

The downstream size effect of increased visual input was evidently not 1:1 in primates; a small increase in visual input to the neocortex appears to have resulted in a proportionally large expansion of visual areas

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in the OPT cortex. This magnification parallels that observed for the LGN, V1, and extrastriate visual areas (*15*, *47*). Because the neocortex is tightly integrated with the rest of the brain, disproportionate expansion of the OPT cortex may have generated a correlated increase in interconnected structures such as the cerebellum (*56*), as expected under models that emphasize the developmental and functional integration of different brain subregions (*51*).

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SUPPLEMENTARY MATERIALS

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Fossil evidence favors a role for vision in the modular evolution of the primate neocortex

Richard F. Kay, Chris L. Organ, Paul E. Morse, Kari L. Allen, and E. Christopher Kirk

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Editor's summary

Expansion of the neocortex is a key differentiator of the primate brain relative to other mammals. Kay *et al*. used fossil primate brain virtual endocasts to understand how the neocortex evolved (see the Perspective by Barton). The authors showed that enhanced visual processing, not expansion of the frontal lobe, drove the major increases in neocortical size. These findings argue against the assumption that frontal lobe enlargement was the primary innovation underlying primate brain evolution. —Mattia Maroso

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